

Scenario: University Course Management System

In this system, a university needs to manage students, courses, faculty members, and enrollments. The system handles course offerings, student registrations, and faculty assignments to courses. It also tracks the grades given by instructors to students for the courses they take.

1. Entities and Attributes

Entity 1: Student

A student represents an individual enrolled in the university.

Attributes:

- Student_ID (Primary Key)
- First_Name
- Last_Name
- Email
- Date_of_Birth
- Enrollment_Date
- Major

Entity 2: Course

A course represents an academic course offered by the university.

Attributes:

- Course_ID (Primary Key)
- Course_Name
- Course_Code
- Department
- Credits
- Semester_Offered

Entity 3: Faculty

A faculty member represents a professor or lecturer teaching the courses.

Attributes:

- Faculty_ID (Primary Key)
- First_Name
- Last_Name
- Email
- Department
- Date_of_Joining
- Office_Number

Entity 4: Enrollment

An enrollment represents the registration of a student in a specific course for a given semester.

Attributes:

- Enrollment_ID (Primary Key)
- Student_ID (Foreign Key)
- Course_ID (Foreign Key)
- Enrollment_Date
- Grade (initially NULL, filled after course completion)

Entity 5: Department

A department represents an academic department within the university.

Attributes:

- Department_ID (Primary Key)
- Department_Name
- Faculty_Count
- Location

2. Relationships

Relationship 1: Student and Enrollment

- Description: A student can enroll in many courses, and each enrollment corresponds to one student.
- Cardinality: One-to-Many (1:N)
- Modality: Mandatory (A student must have at least one enrollment.)

Relationship 2: Course and Enrollment

- Description: A course can have many students enrolled, and a student can enroll in multiple courses.
- Cardinality: Many-to-Many (M:N)
- Modality: Optional (A course can have zero students enrolled, but an enrollment must be associated with a student.)

To resolve this many-to-many relationship, we use the Enrollment associative entity.

Relationship 3: Faculty and Course

- Description: A faculty member can teach many courses, but each course is taught by exactly one faculty member.
- Cardinality: One-to-Many (1:N)
- Modality: Optional (A course may not always have a faculty member assigned yet.)

Relationship 4: Department and Faculty

- Description: A department can have multiple faculty members, but a faculty member is assigned to exactly one department.
- Cardinality: One-to-Many (1:N)
- Modality: Mandatory (Each faculty member must belong to one department.)

3. ER Diagram Representation:

Now let's summarize the entities, relationships, and cardinality in a textual format:

1. Student (1) → (M) Enrollment
 - A student must have at least one enrollment, and a student can enroll in many courses.
2. Course (M) → (M) Enrollment
 - A course can have multiple students enrolled, and a student can enroll in many courses. This is managed by the Enrollment entity.
3. Faculty (1) → (M) Course
 - A faculty member can teach multiple courses, but each course is taught by exactly one faculty member.
4. Department (1) → (M) Faculty
 - A department can have multiple faculty members, and each faculty member is assigned to one department.

4. ER Diagram:

Representation of the entities, relationships, and cardinality:

Entities:

- Student (Student_ID, First_Name, Last_Name, Email, Date_of_Birth, Enrollment_Date, Major)
- Course (Course_ID, Course_Name, Course_Code, Department, Credits, Semester_Offered)
- Faculty (Faculty_ID, First_Name, Last_Name, Email, Department, Date_of_Joining, Office_Number)
- Enrollment (Enrollment_ID, Student_ID, Course_ID, Enrollment_Date, Grade)
- Department (Department_ID, Department_Name, Faculty_Count, Location)

Relationships:

- Student (1) → (M) Enrollment
- Course (M) → (M) Enrollment
- Faculty (1) → (M) Course
- Department (1) → (M) Faculty

5. Cardinality and Modality Summary:

Relationship	Cardinality	Modality
Student → Enrollment	1:N	Mandatory (Student must enroll in at least one course)
Course → Enrollment	M:N	Optional (A course can have no students enrolled, but an enrollment must have a student)
Faculty → Course	1:N	Optional (A course may not have a faculty member assigned yet)
Department → Faculty	1:N	Mandatory (Faculty must belong to a department)

Class Diagram for the University Course Management System

1. Student

- Attributes:
 - studentID: String
 - firstName: String
 - lastName: String
 - email: String
 - dateOfBirth: Date
 - enrollmentDate: Date
 - major: String
- Methods:
 - enroll(course: Course): Enrollment
 - viewCourses(): List<Course>
 - getGrade(course: Course): Grade

2. Course

- Attributes:
 - courseID: String
 - courseName: String
 - courseCode: String
 - department: String
 - credits: int
 - semesterOffered: String
- Methods:
 - addStudent(student: Student): Enrollment
 - getFaculty(): Faculty
 - getEnrolledStudents(): List<Student>

3. Faculty

- Attributes:
 - facultyID: String
 - firstName: String
 - lastName: String
 - email: String
 - department: String
 - dateOfJoining: Date
 - officeNumber: String
- Methods:
 - assignCourse(course: Course)
 - gradeStudent(course: Course, student: Student, grade: Grade)

4. Enrollment

- Attributes:
 - enrollmentID: String
 - student: Student
 - course: Course
 - enrollmentDate: Date
 - grade: Grade
- Methods:
 - assignGrade(grade: Grade)

5. Department

- Attributes:
 - departmentID: String
 - departmentName: String
 - facultyCount: int
 - location: String
- Methods:
 - getFacultyList(): List<Faculty>
 - getCoursesOffered(): List<Course>

6. Grade (A supporting class for storing grade information)

- Attributes:
 - gradeID: String
 - gradeValue: String (A value like "A", "B+", "C", etc.)
 - gradeDate: Date
- Methods:
 - getGradeDescription(): Returns a description of the grade (e.g., "Excellent", "Good").

Relationships Between Classes:

1. Student - Enrollment:
 - A Student can have many Enrollment objects, as they can enroll in multiple courses.
 - One Enrollment is associated with one Student and one Course.
 - Cardinality: One-to-Many (1:N)
2. Course - Enrollment:
 - A Course can have many Enrollment objects (i.e., multiple students can enroll in the same course).
 - Each Enrollment links exactly one Course and one Student.
 - Cardinality: One-to-Many (1:N)
3. Course - Faculty:
 - A Course is taught by exactly one Faculty member, but a Faculty member can teach many Course objects.
 - Cardinality: One-to-Many (1:N)
4. Department - Faculty:
 - A Department can have many Faculty members, but each Faculty member is assigned to only one Department.
 - Cardinality: One-to-Many (1:N)
5. Enrollment - Grade:
 - An Enrollment can have one Grade, which indicates how a student performed in a particular course.
 - Cardinality: One-to-One (1:1)